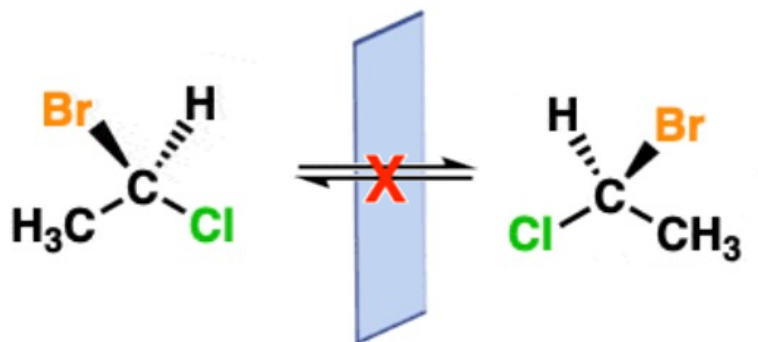


Conformers and Applications of Stereoisomerism

Stereoisomers: Interconversion is not possible without breaking bonds

How to convert R to S?

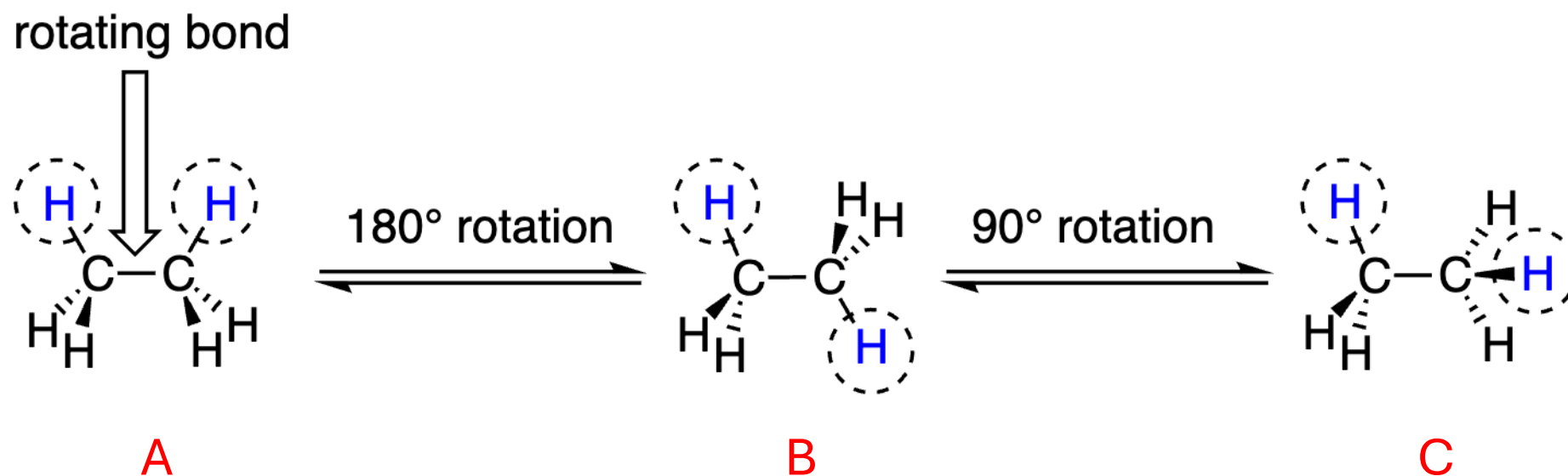


Strength of sigma C-C ~90KCal/mole

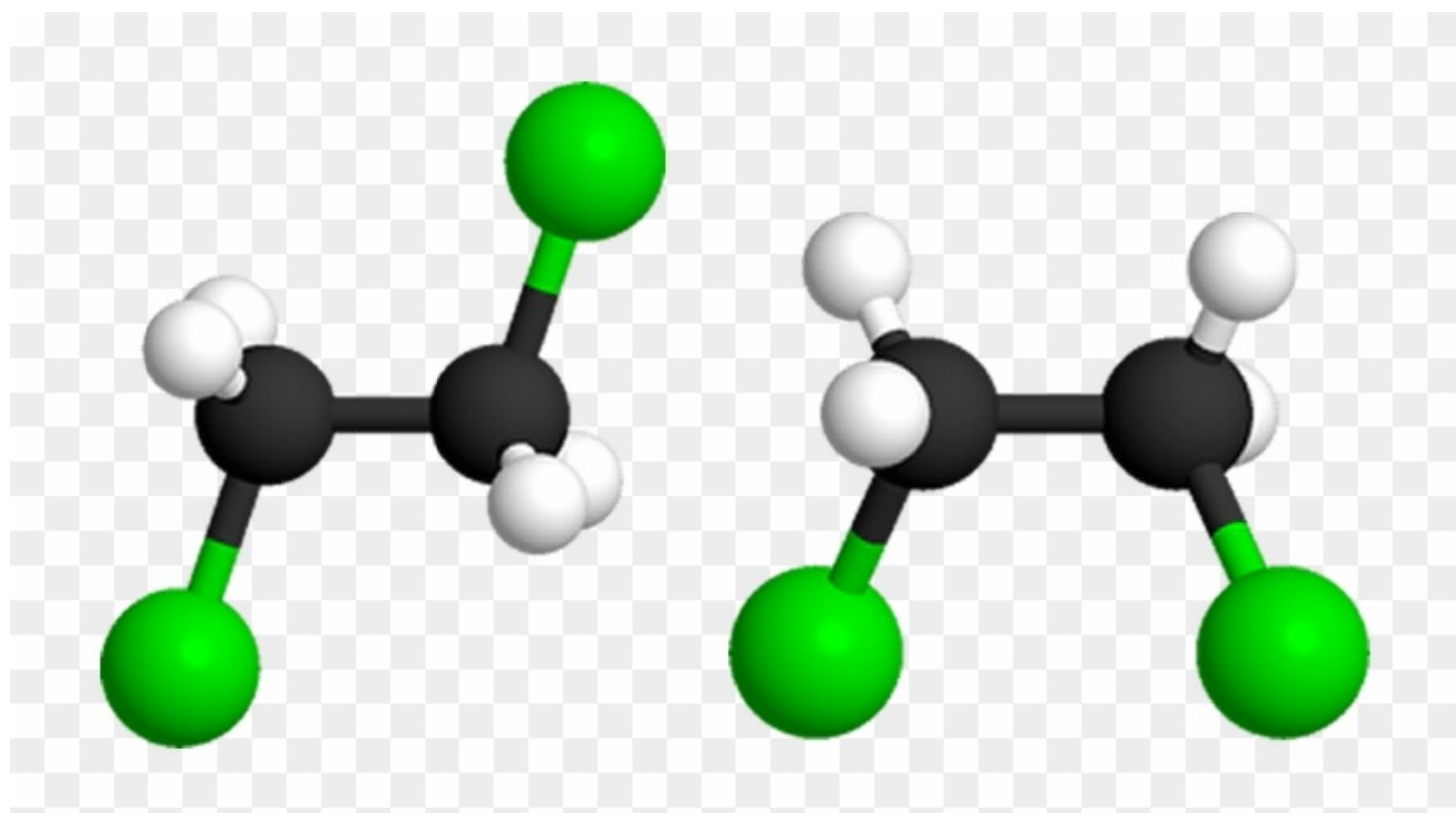
How to convert cis to trans?



Strength of pi C-C ~60KCal/mole



Free rotation along C-C bonds
Can there be isomers **A**, **B** and **C** for ethane?



Is energy required to rotate C-C bonds? Why?

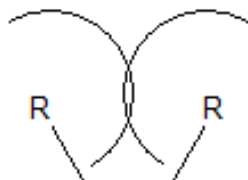
What amount of energy are we talking about?

The Concept of Strain

*Picture yourself moving through a crowded street vs an empty street
Where do you think it will be easier to move?*



Steric Clash – Bulky groups



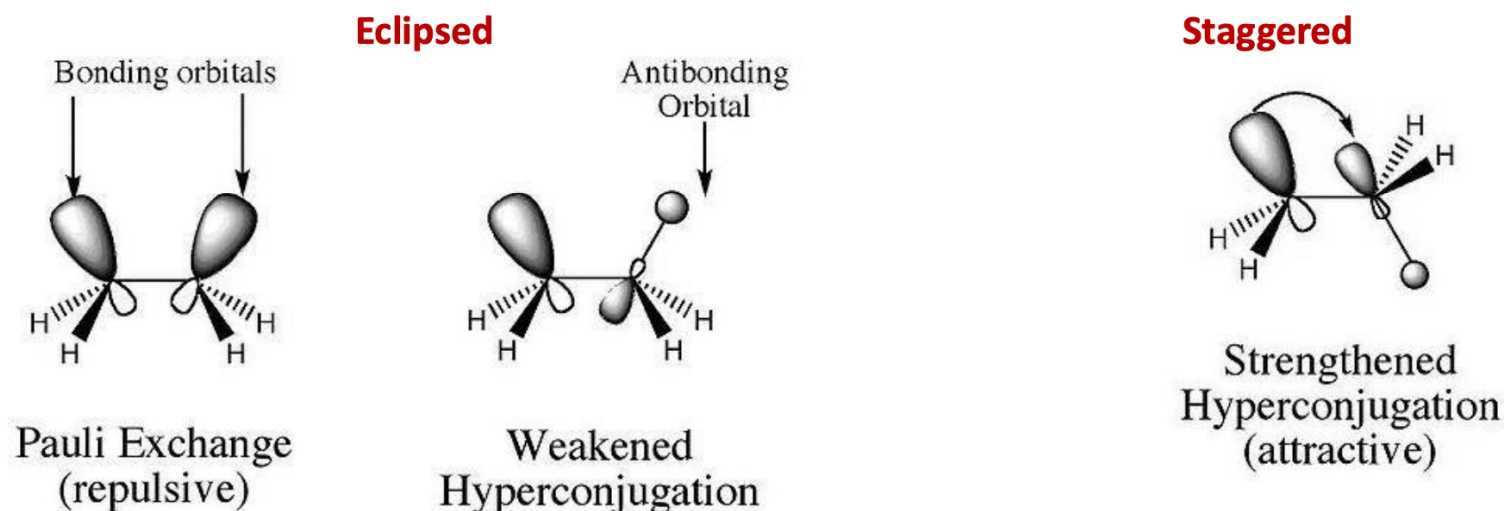
Strain in Organic Molecules

Steric (non-bonded interactions) – Destabilization due to the repulsion between the electron clouds of atoms or groups. Groups compete to occupy common space. Strain due to close contact of atoms separated by four or more bonds

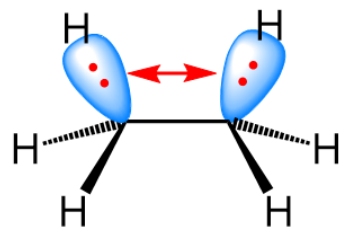
Torsional – Strain resulting from eclipsing interactions between bonds on adjacent carbon atoms. Actual reasons are highly debated.

Angular – Destabilization due to distortion of a bond angle from its optimal value

Possible Reasons for Torsional Strain

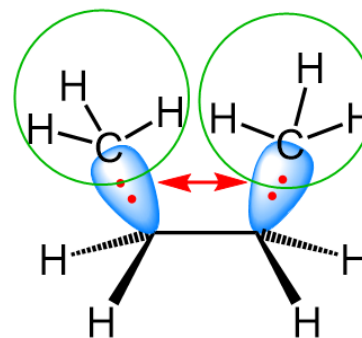


Torsional and Steric Interactions



Torsional Strain

Repulsion between electrons in the bond destabilizes the eclipsed conformation of ethane.

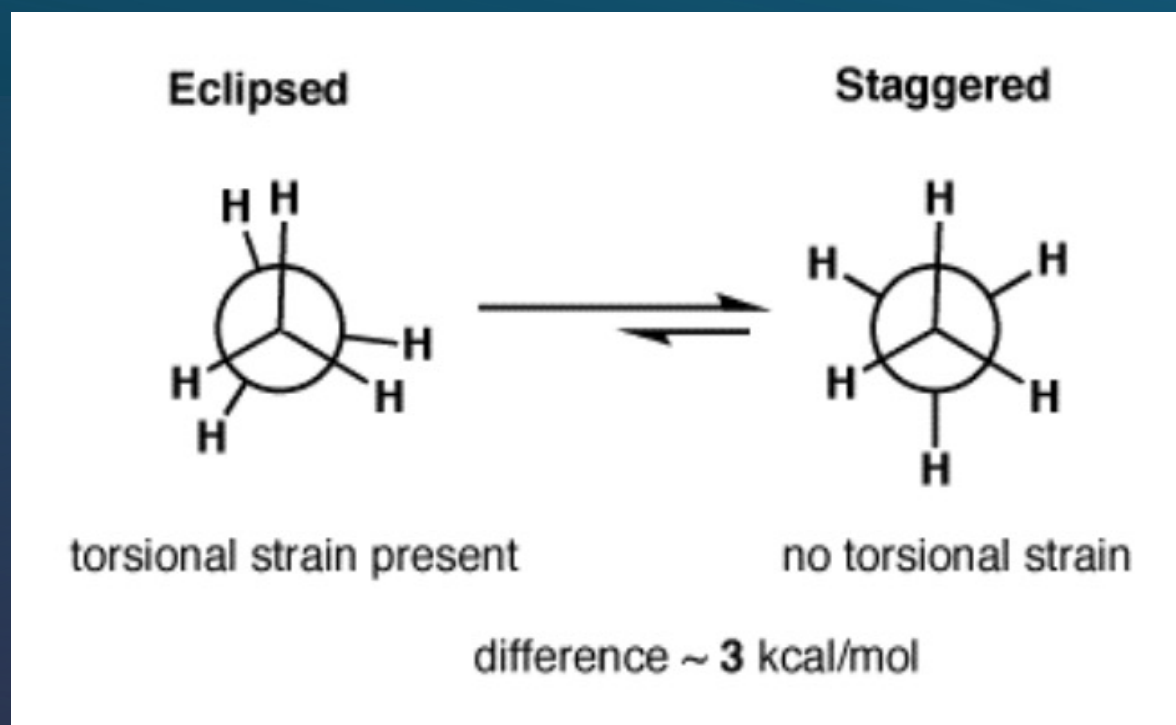


Steric Strain

Interaction caused by large atoms/groups getting in each other's way.

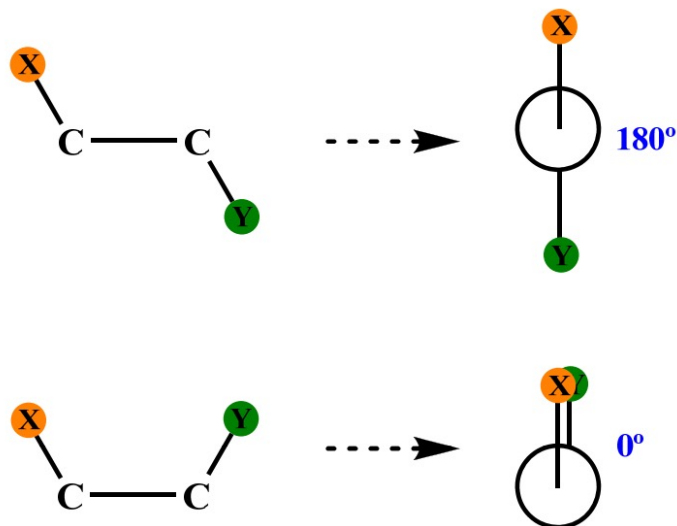
- *Hydrogens are too small and it is the **bonding electrons that repel** each other (**torsional strain**).*
- *Methyl **groups are large** and, in addition to this torsional strain between each eclipsing bond, there is also a **steric interaction** between them.*

Strain governs equilibrium



The two "conformations" will have slightly different energy states

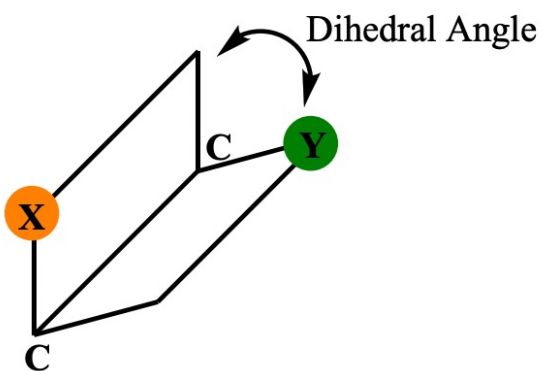
Dihedral Angle



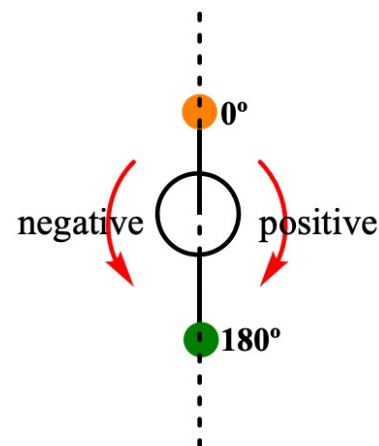
A dihedral angle is the angle between two intersecting planes or half-planes

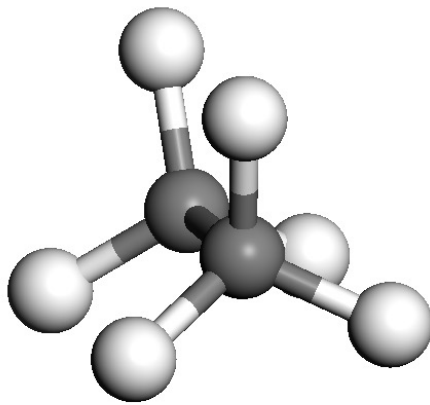
A parameter used to define different conformers

The angle defined by X-C-C and C-C-Y plane is termed as dihedral angle.



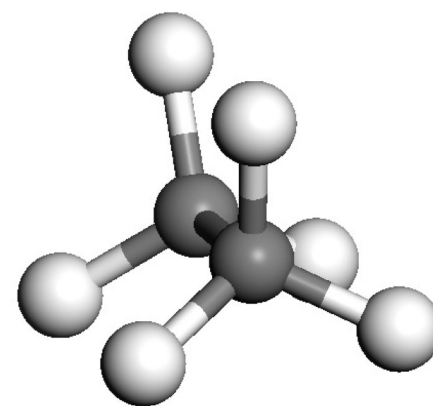
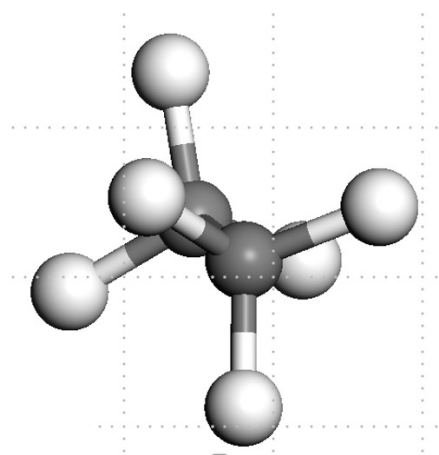
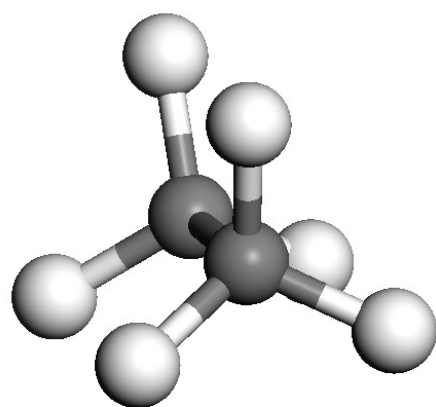
Torsional angle is an alternative term used for dihedral angle. While dihedral angle is measured from 0 - 360° , torsional angle is measured from 0 - 180° , as a positive value in the clockwise direction and a negative value in the anti-clockwise direction.





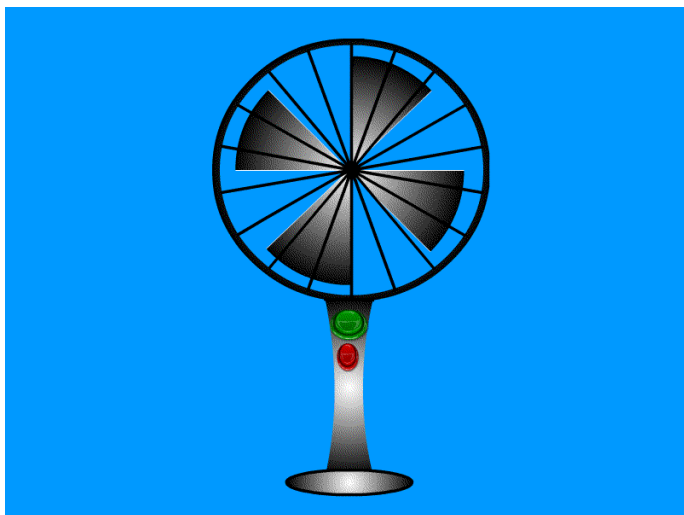
There are infinite number of conformers

Each distinct dihedral angle (θ) defines a distinct conformation



Analogy of conformer interconversion: Rotating blades of a fan

Fan A



Fan B



Most conformers interconvert rapidly at room temperature (like blades in FAN A)
Difficult to detect single conformers (a single blade of the fan) at room temperature because of fast rotation of single bonds (high speed of the fan)
Most of the detection methods provide a time-averaged structure
Lowering the temperature can slow down the rotation and allow us to detect single conformers (just like turning down a fan speed to see the individual blades)

How can we detect single conformers?

Whether we will be able to detect a single conformer or a time-averaged structure will depend on the **rate of interconversion (C-C bond rotation)** and **the timescale of the method** used

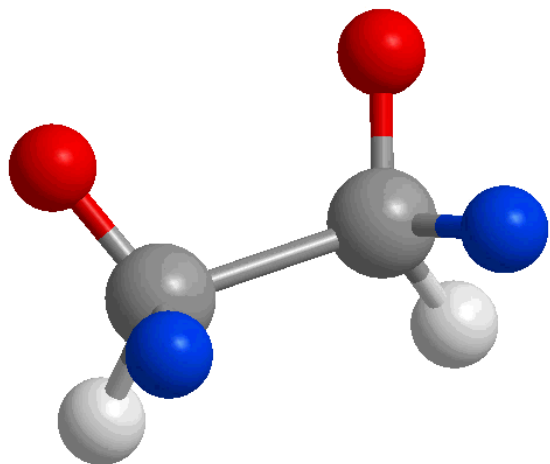
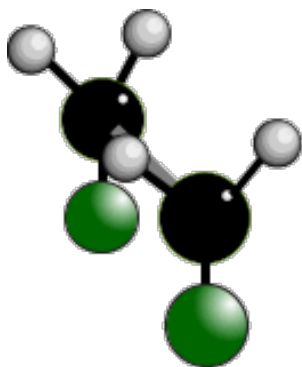
Rapid interconversion will lead to detection of a time-averaged structure at room temperature

Lowering the temperature will slow down bond rotation and enable detection of single conformers

Cryogenic temperature (temperature of liquid nitrogen) = 77 K or -196°C

At what temperature, structural resolution will be obtained will depend on the energy difference (gap) between the conformers (rotational energy barrier; discussed in upcoming slides)

However, some detection methods operate on an extremely fast timescale and can be used to detect conformers even at room temperature



A few methods to detect conformers

Technique	Typical Timescale	What It Measures (Basic Principle)	What You See for Conformers
IR spectroscopy	$\sim 10^{-13} - 10^{-14}$ s	Molecular vibrational energy levels (bond stretching, bending)	Usually time-averaged conformations (fast rotation)
Raman spectroscopy	$\sim 10^{-13} - 10^{-14}$ s	Change in polarizability during molecular vibrations	Time-averaged; sensitive to symmetry
NMR spectroscopy	$\sim 10^{-3} - 10^{-1}$ s	Magnetic environment of nuclei (chemical shifts, coupling)	Strongly time-averaged at room temperature
Electron diffraction (gas phase)	$\sim 10^{-15}$ s	Scattering of electrons by atomic positions	Can resolve instantaneous geometries
Microwave spectroscopy	$\sim 10^{-9} - 10^{-6}$ s	Rotational energy levels of molecules	Can distinguish individual conformers

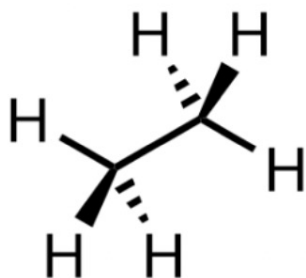
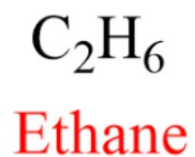
Timescale of C-C single bond rotation in alkanes

Different conformers and the energy gaps

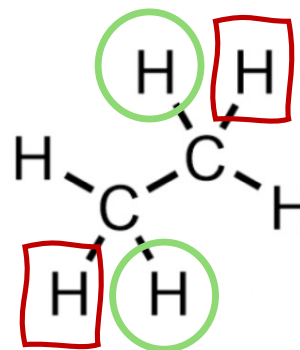
How to draw conformers?

Sawhorse Projection

Representing a 3D structure on a 2D plane



3D



2D

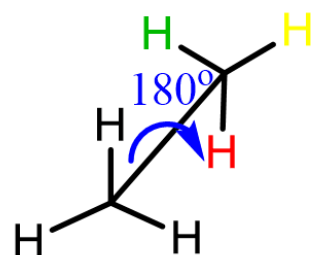
Replace hash and dash bonds with solid bonds

Green: Below the plane
Red: Above the plane

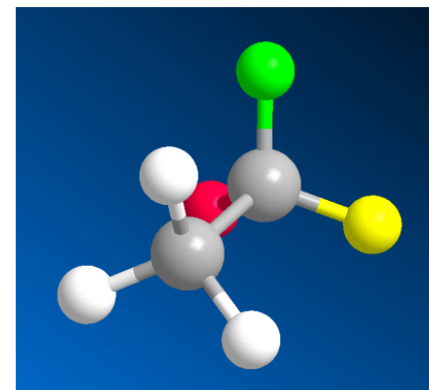
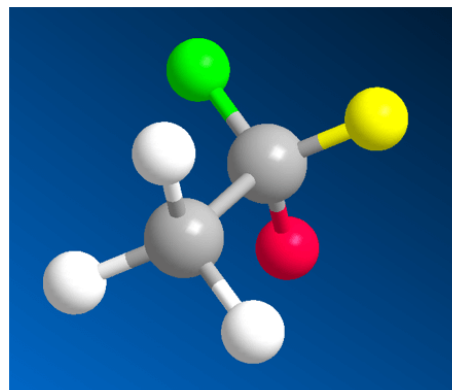
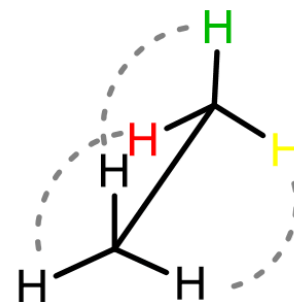


Zig-zag structure
Side view of the molecule

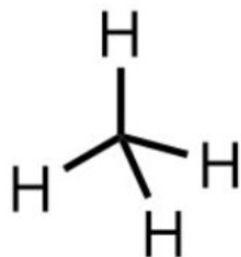
Staggered



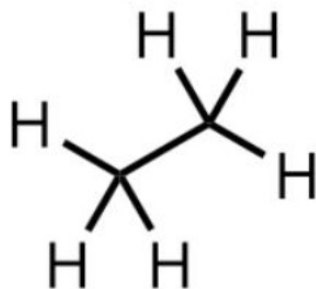
Eclipsed



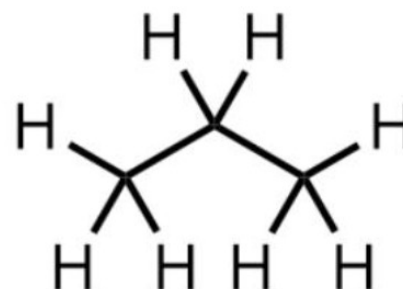
Sawhorse Projection of Acyclic Molecules



Methane



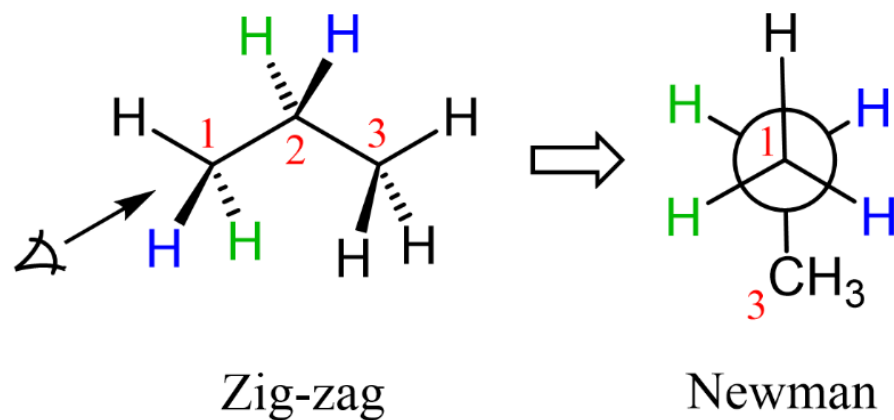
Ethane



Propane

Newman's Projection

Representing a 3D structure on a 2D plane

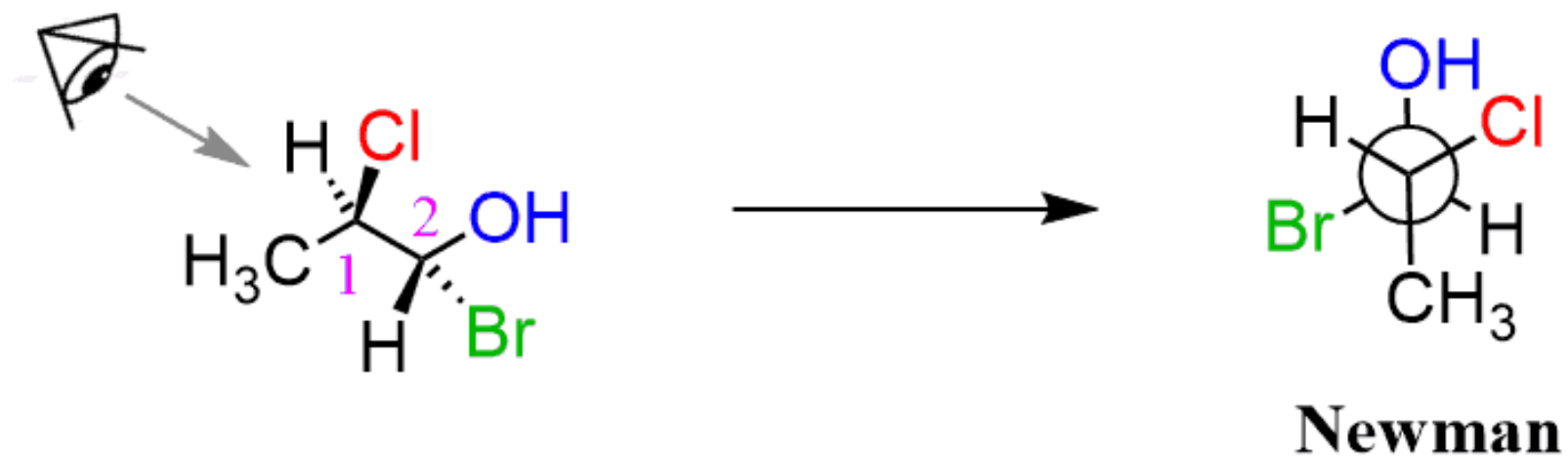


Propane, C₃H₈

For drawing Newmann projection, the direction of view must be mentioned

Newman's Projection

Representing a 3D structure on a 2D plane

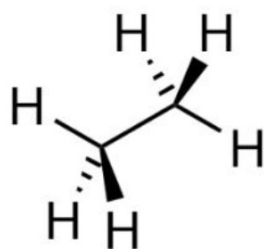
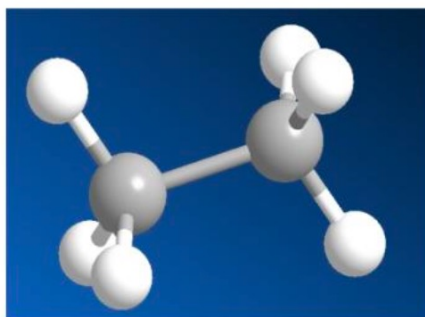


@jamesashchem

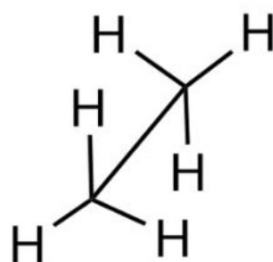
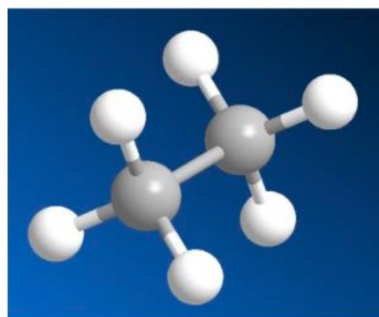


Newman's Projection

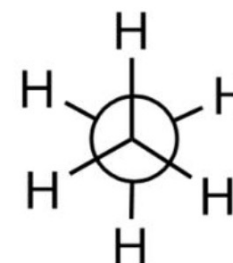
Representing a 3D structure on a 2D plane



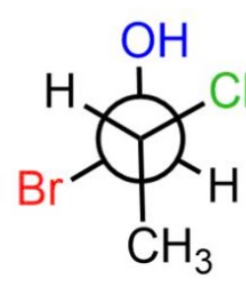
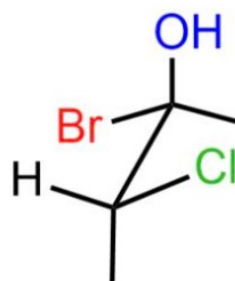
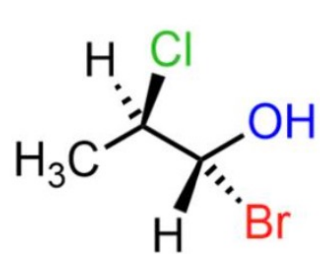
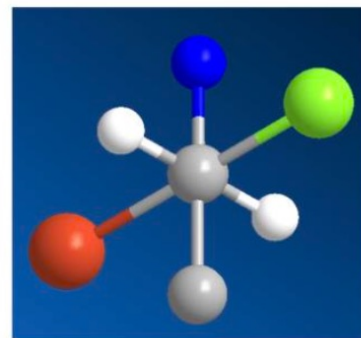
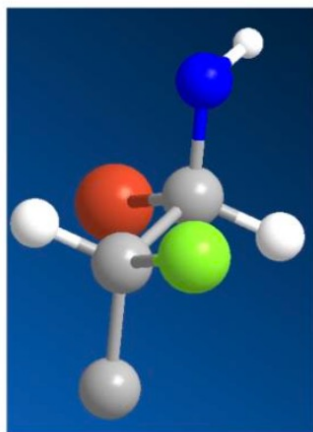
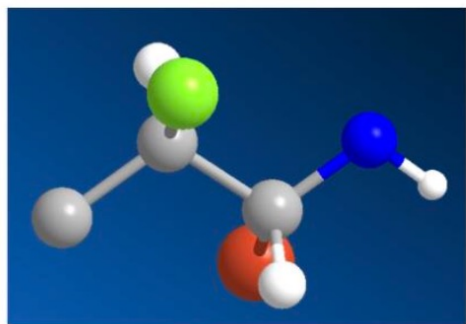
Bond-line



Sawhorse



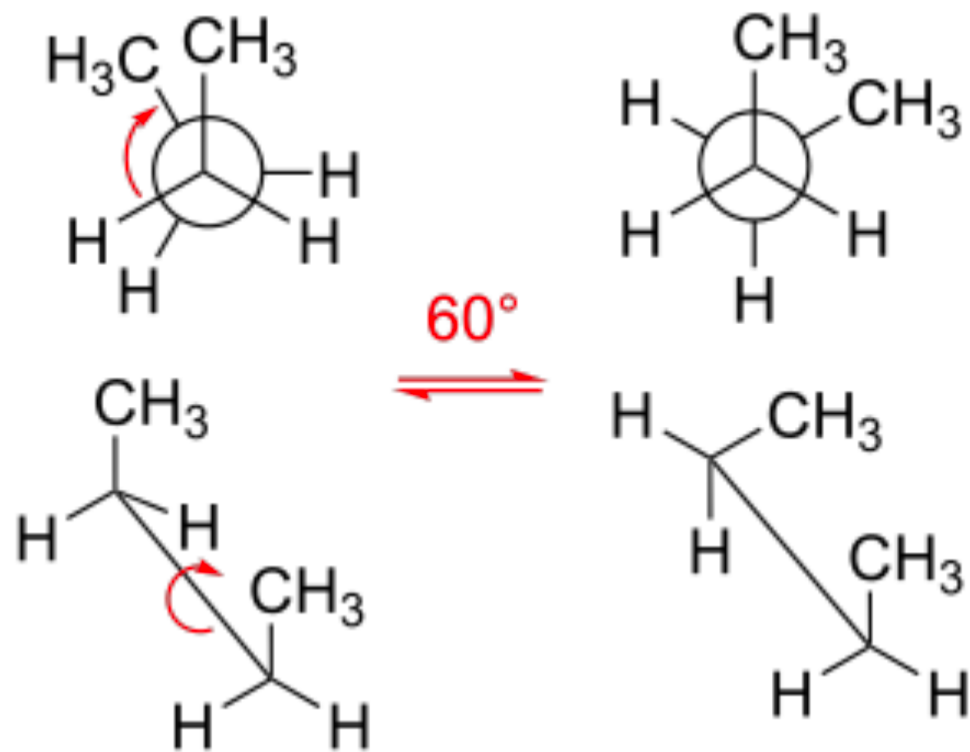
Newman



Bond-line

Sawhorse

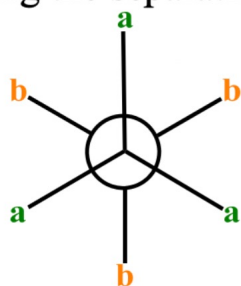
Newman



Conformers of butane shown in Newman and Sawhorse projections

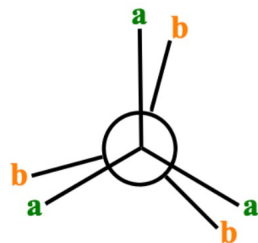
Conformations

Staggered - A low energy conformation where the bonds on adjacent atoms bisect each other (60° dihedral angle), maximizing the separation.



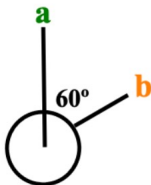
Staggered or anti conformation

Eclipsed - A high energy conformation where the bonds on adjacent atoms are aligned with each other (0° dihedral angle).



“Full eclipsed” vs “Partial Eclipsed”

Gauche - Description given to two substituents attached to adjacent atoms when their bonds are at 60° with respect to each other.



Gauche or skew conformation

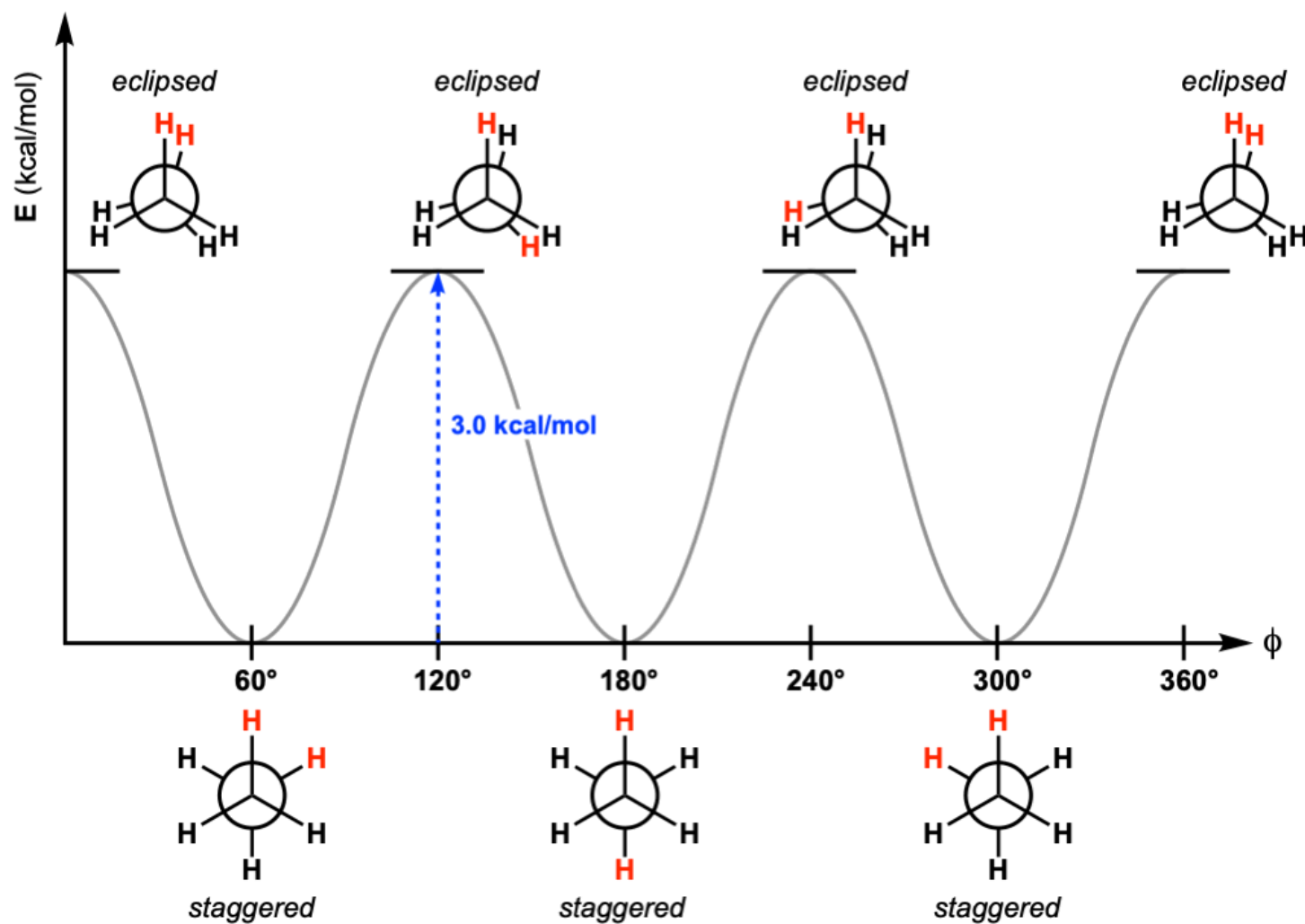
Draw Sawhorse and Newmann projections (C2-C3 bond) of n-pentane



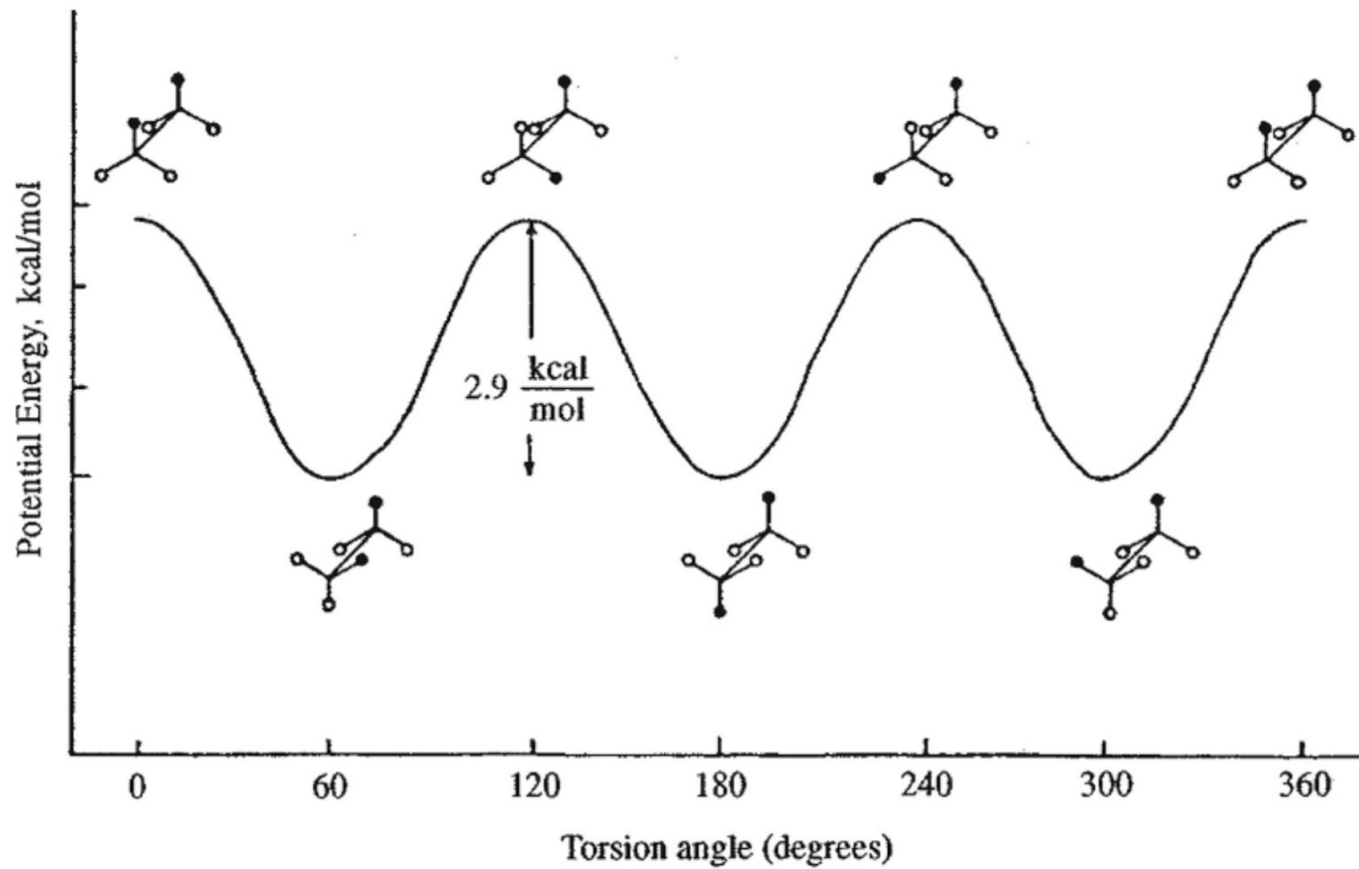
Staggered, Gauche, Eclipse conformations

Graphing the rotational barrier in ethane (C_2H_6) as a function of dihedral angle

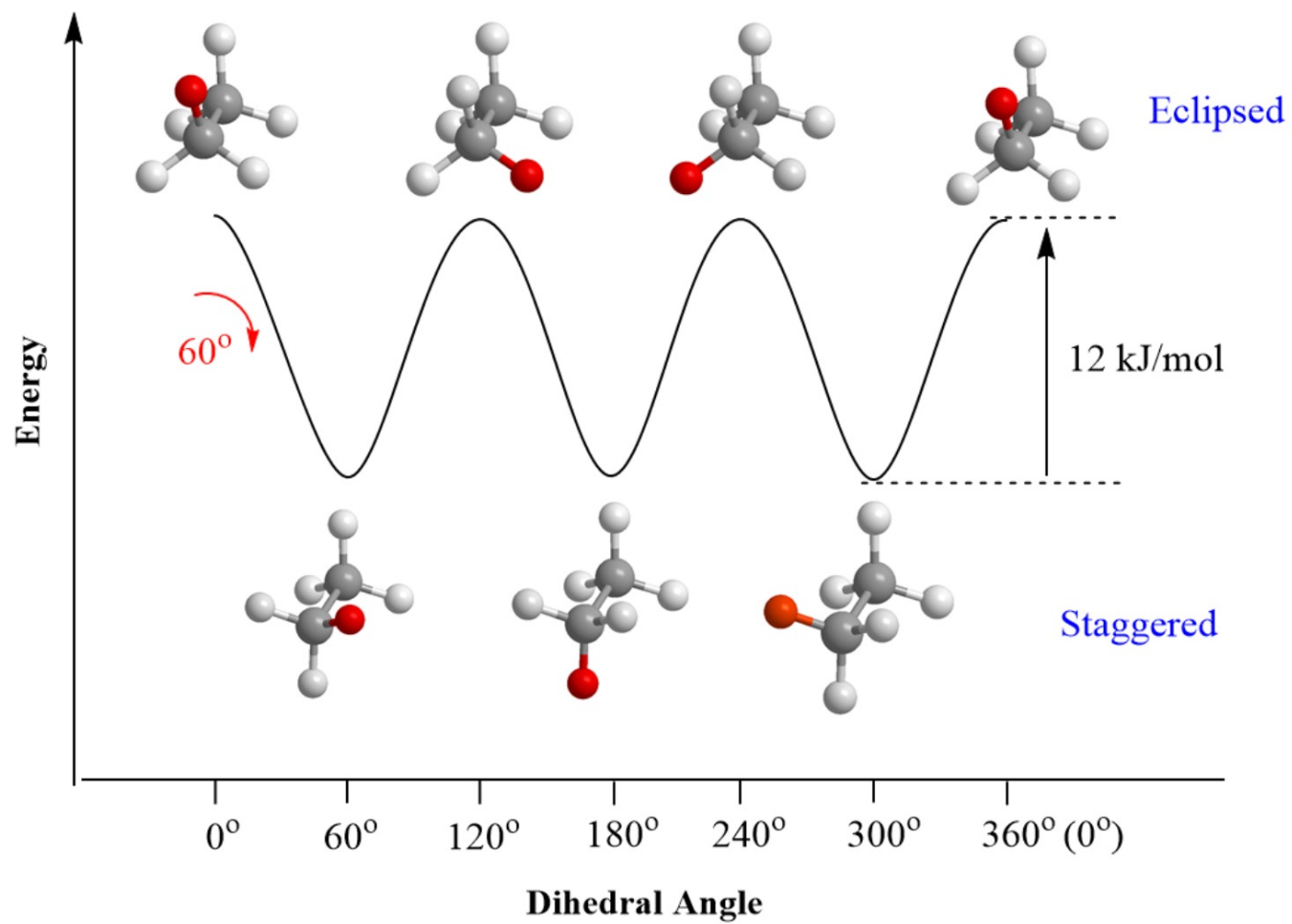
The barrier to rotation in ethane is about 3.0 kcal/mol.



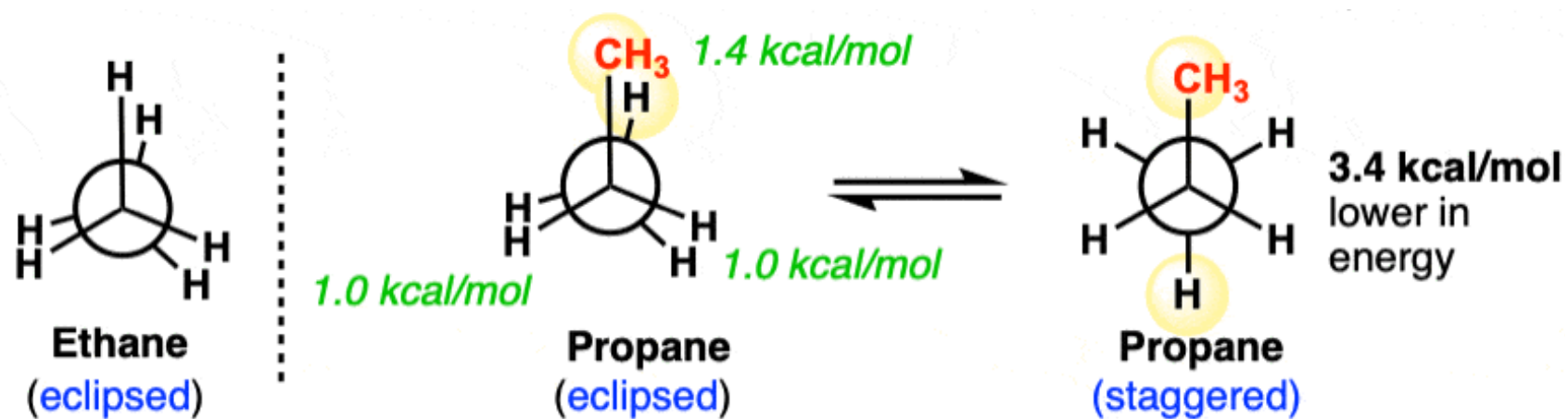
Sawhorse projections



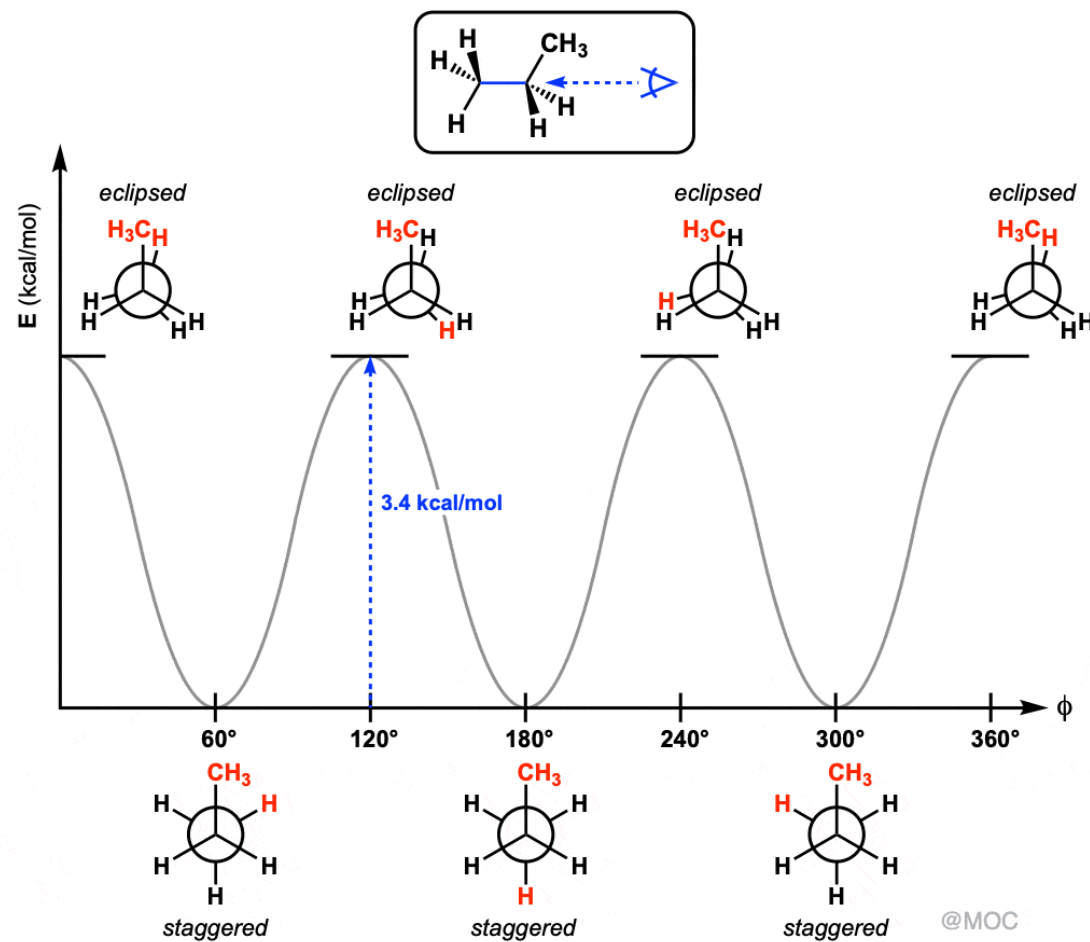
The Conformational Analysis of Ethane



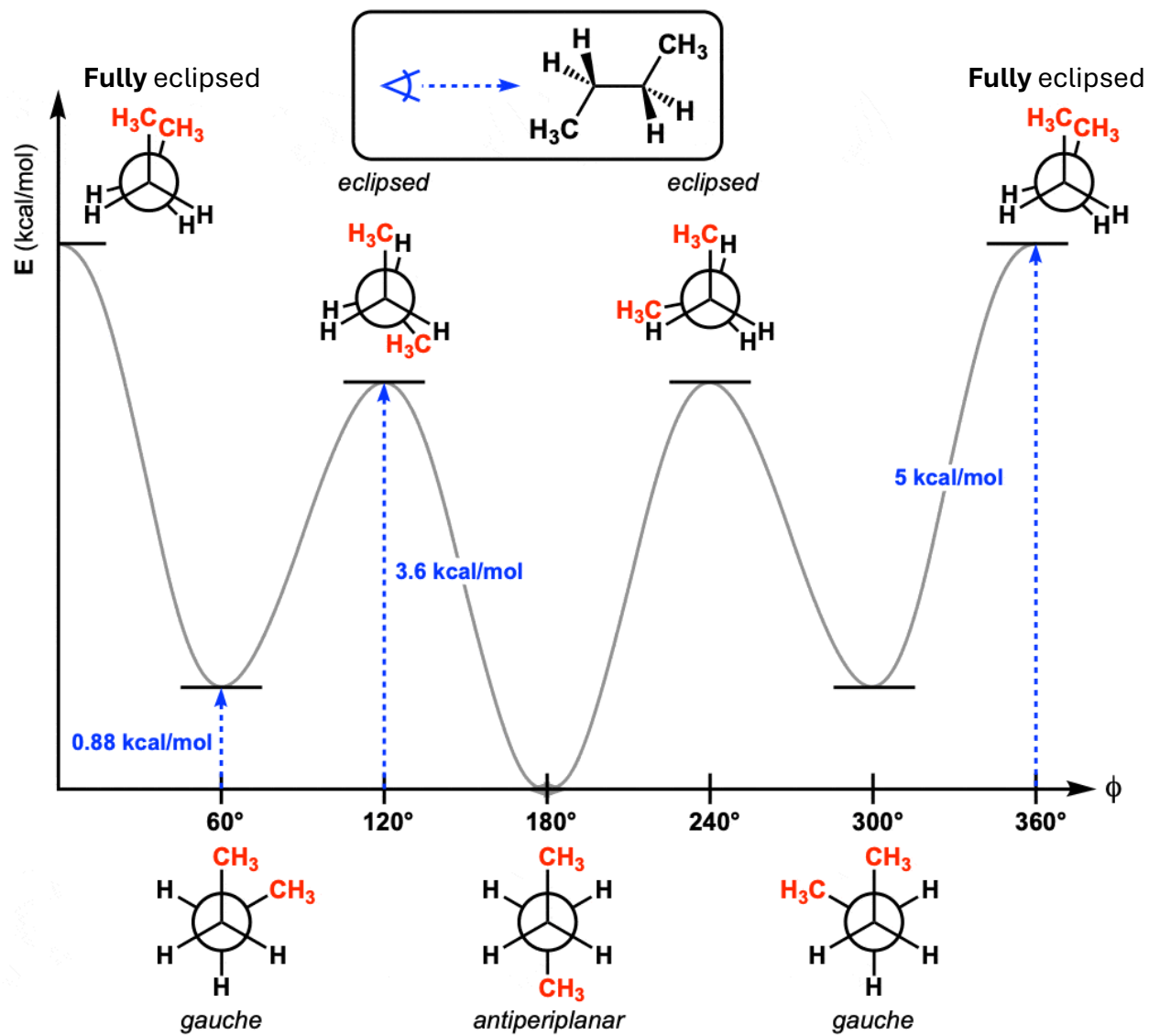
Potential Energy Diagram for Conformers of Propane



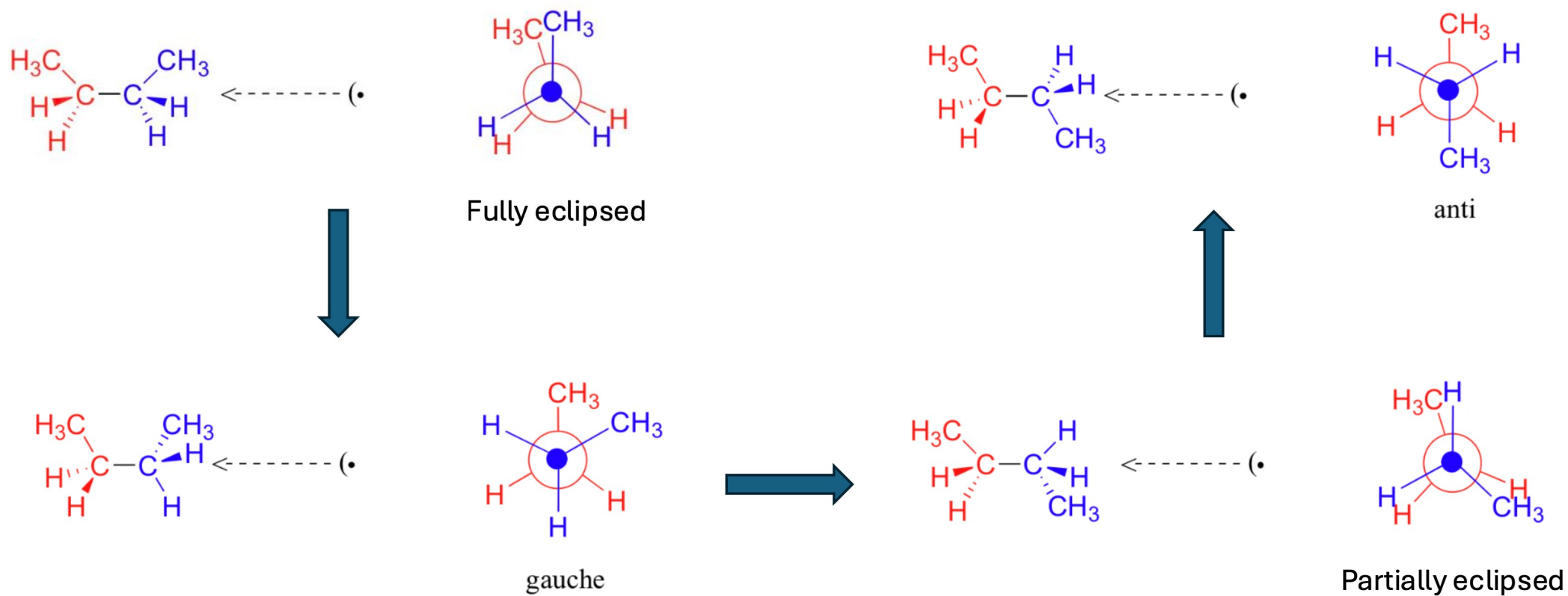
Energy Diagram for Rotation Of The C—C Bond In Propane



Graph Of Strain Energy vs. Dihedral Angle For Butane (Along C₂-C₃ Bond)



Newmann projections from 3D structures



In general: Anti/ Staggered conformation is the most stable

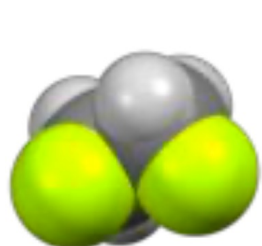
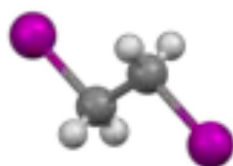
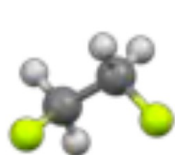
In certain cases, gauche or skew conformation can be the most stable

In rare cases, eclipse conformation can be the most stable

Gauche Effect

The gauche conformer is more stable than the anti conformer

Comparison of most stable conformer in 1,2-difluoroethane vs 1,2-diiodoethane

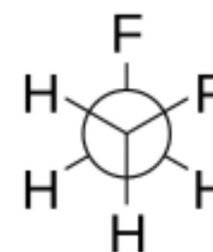
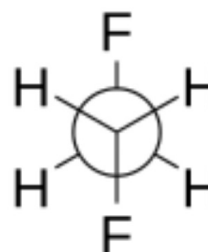
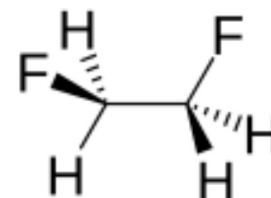
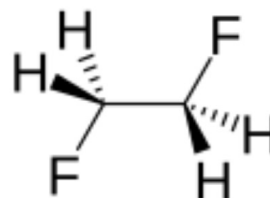


Gauche

Anti

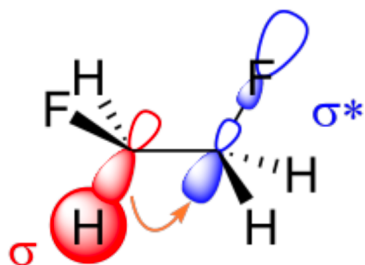
F = yellow
I = pink
C = dark grey
H = light grey

1,2- Difluoroethane prefers gauche conformation
While 1,2-Diiodoethane prefers anti conformation.
Can you explain why?



Why is the gauche more stable than anti?

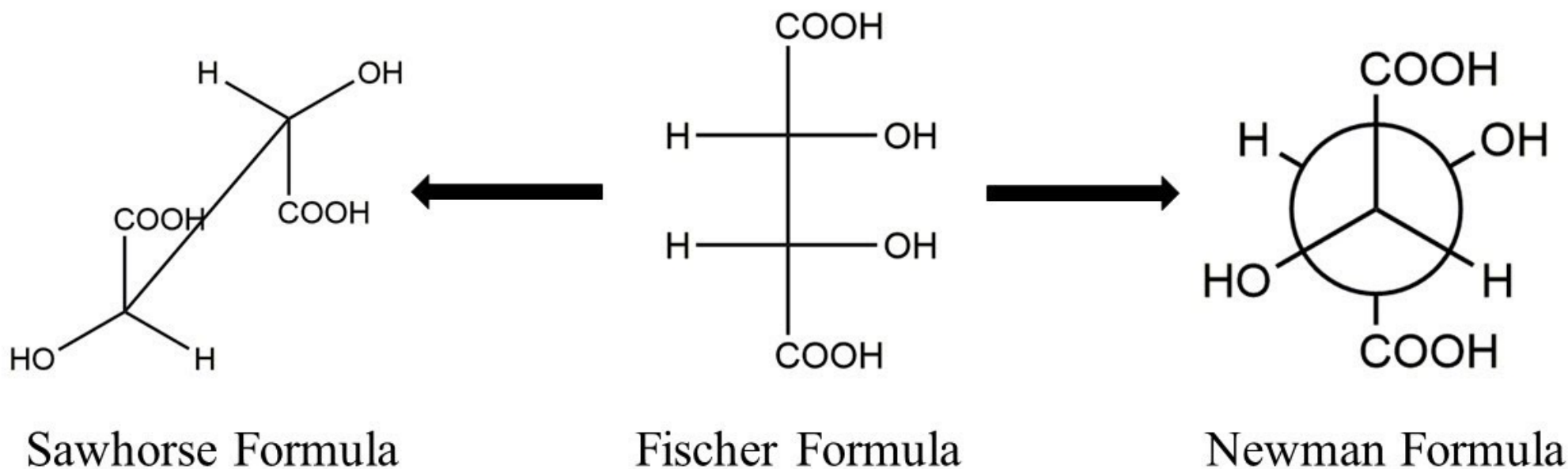
The answer is because of hyperconjugation gauche conformer is more stabilized

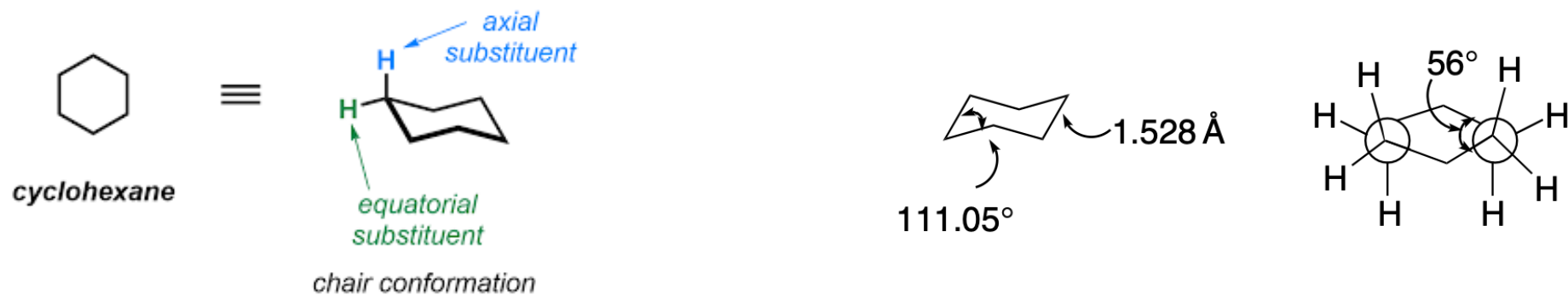


Hyperconjugation model for explaining the gauche effect in 1,2-difluoroethane

the donation of electron density from the C-H σ bonding orbital to the C-F σ^* antibonding orbital is considered the source of stabilization in the gauche isomer

Fischer formula to Sawhorse and Newman Formula



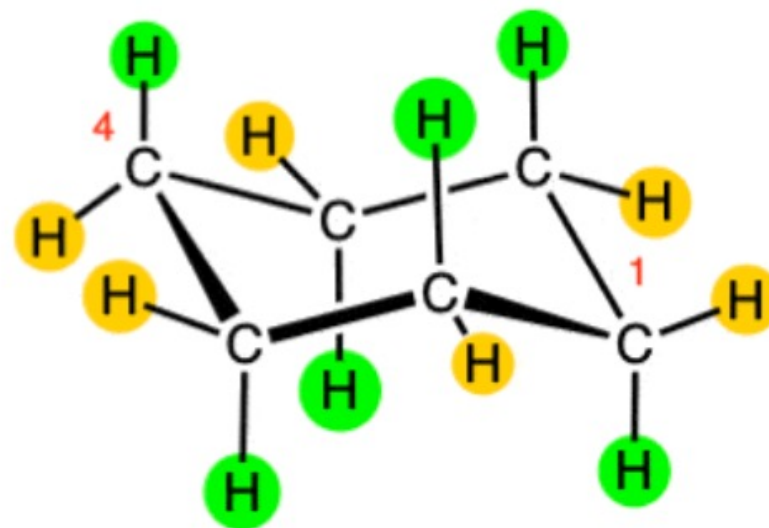


Conformations in Cyclic Molecules

The “Chair” Conformation of Cyclohexane

- **no** angle strain (all bonds 109°)
- **minimal** torsional strain since all C–C bonds are staggered

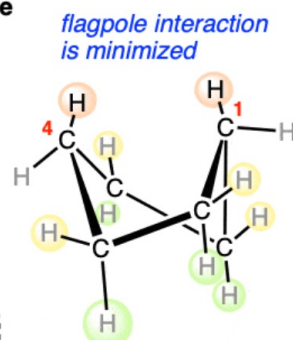
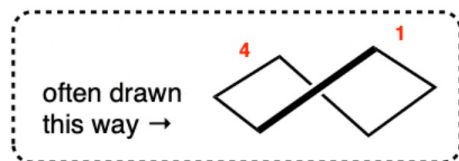
Ring Strain: **0 kcal/mol**



The "Twist-Boat" Conformation of Cyclohexane

- relief of **Van Der Waals** strain through **twisting**
- minimal **angle strain**
- **torsional strain** (some eclipsed C—C bonds)

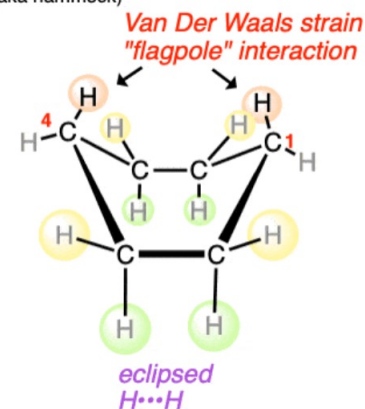
Ring Strain **5.5 kcal/mol**



The "Boat" Conformation of Cyclohexane (aka hammock)

- minimal **angle strain**
- some **torsional strain** (two C—C bonds fully eclipsed, two more partially eclipsed)
- Van Der Waals strain due to close contact between the C—H bonds on C₁ and C₄ ("flagpole" interaction)

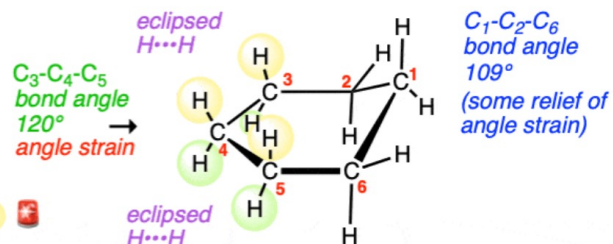
Ring strain **7.0 kcal/mol**



The "Half-Chair" Conformation of Cyclohexane

- some **angle strain**
120° in some places,
109° in others
- some **torsional strain**
(two C—C bonds fully eclipsed, two more partially eclipsed)

Ring strain **10.8 kcal/mol**

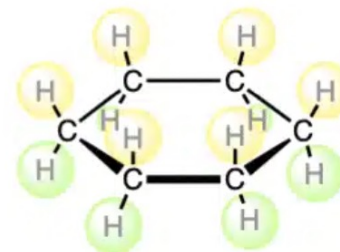


Flat Cyclohexane Is Highly Strained

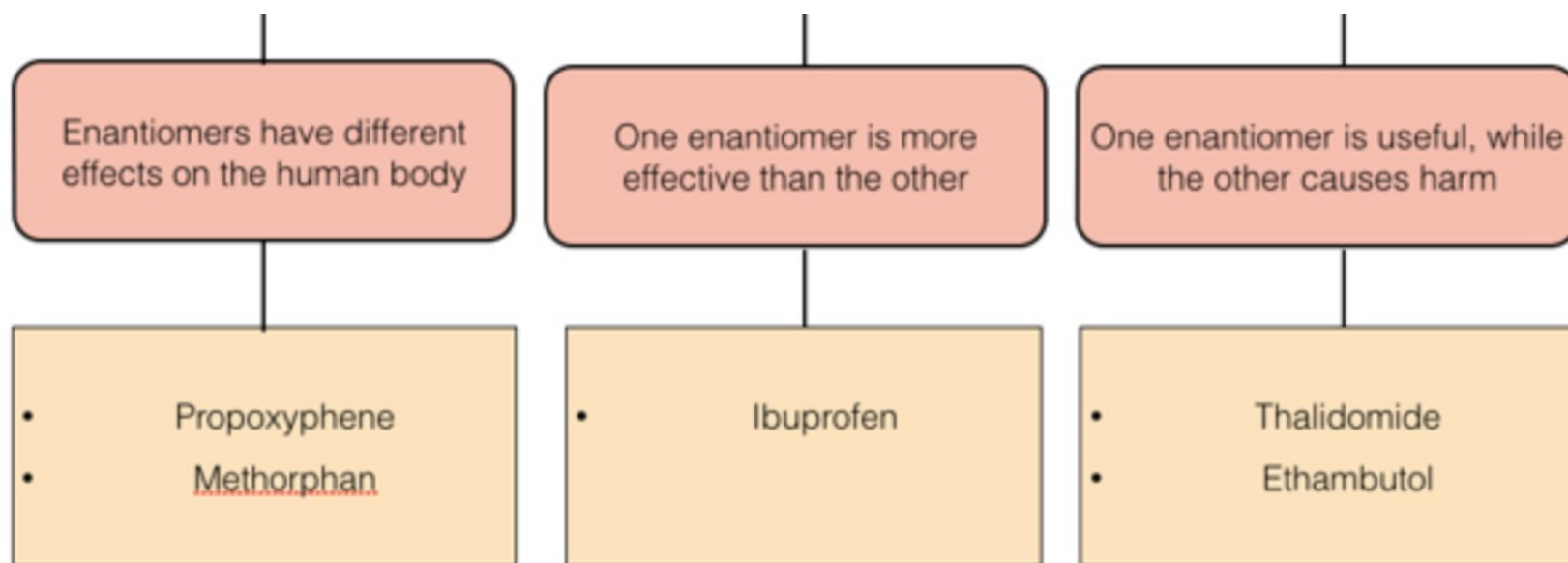
"Flat" cyclohexane

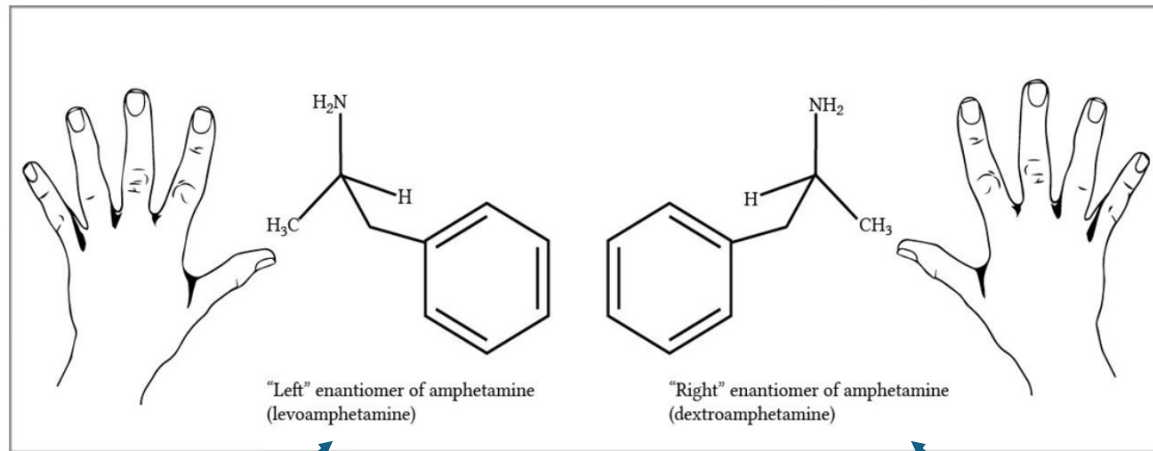
- **angle strain**
(all internal C—C bonds 120°)
- **torsional strain**
(all C—C bonds eclipsed)

Ring strain **> 20 kcal/mol**



Applications of stereochemistry: Pharmaceuticals

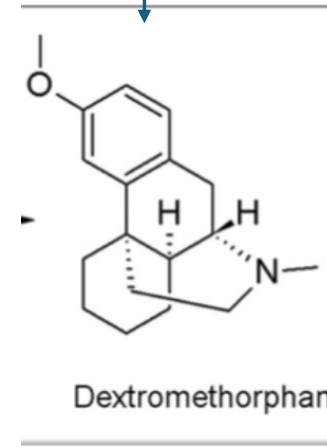
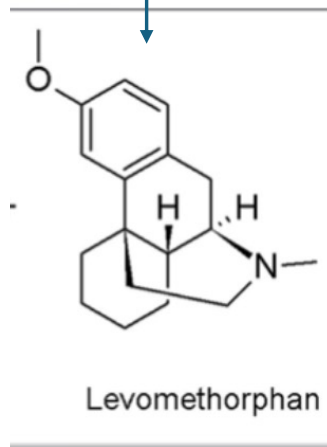




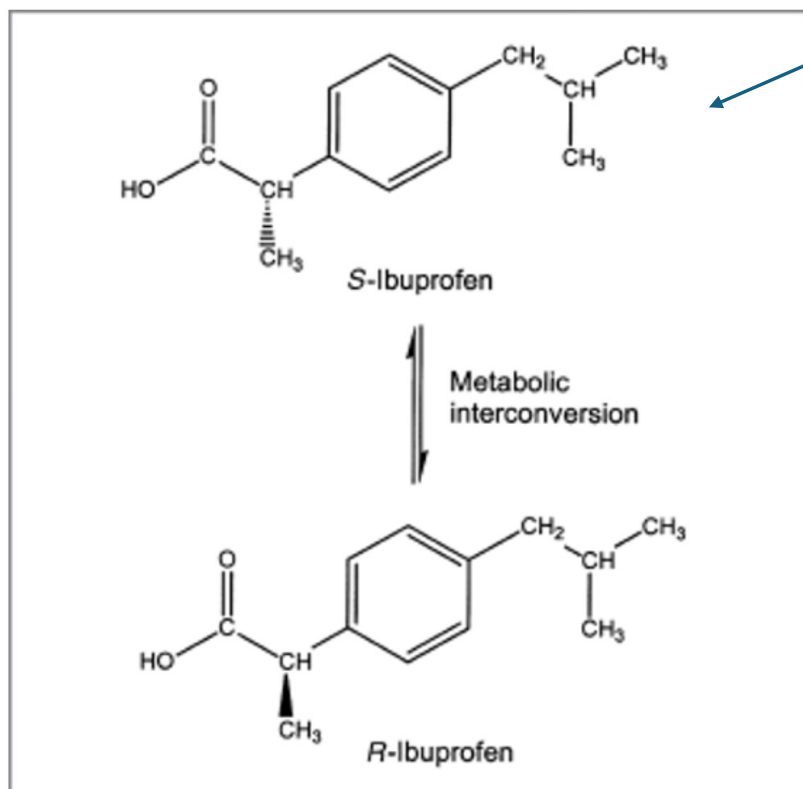
Analgesic (pain killer)

Antitussive (relief coughs)

Opioid analgesic
(morphine)

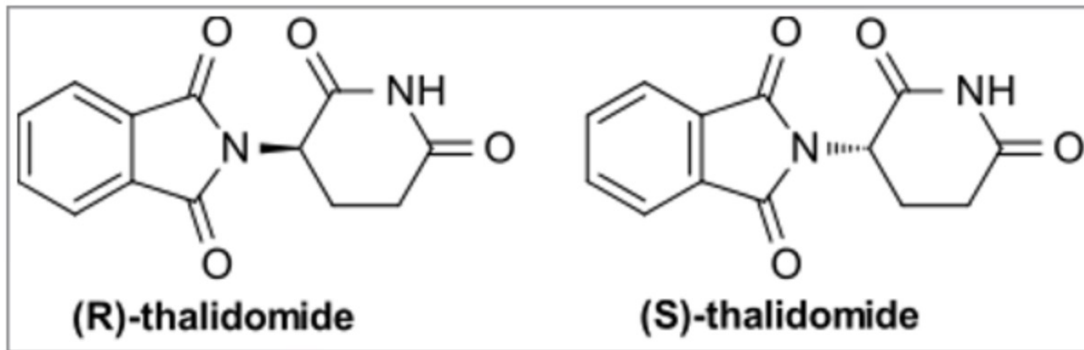


“Active drug”



Ibuprofen is a non-steroidal anti-inflammatory drug that is used to treat conditions such as headaches and muscle pains. Ibuprofen exists as 2 enantiomers— the R(-) ibuprofen and the S(+) ibuprofen. Of the two enantiomers, S-ibuprofen is a more potent inhibitor of the cyclooxygenase enzyme than R-ibuprofen , and hence is more effective in eliciting a response

Infamous drug: The Thalidomide disaster



↑

Sedative (used to treat morning sickness in pregnant women)

↑

teratogenic

Isomers interconvert in the body



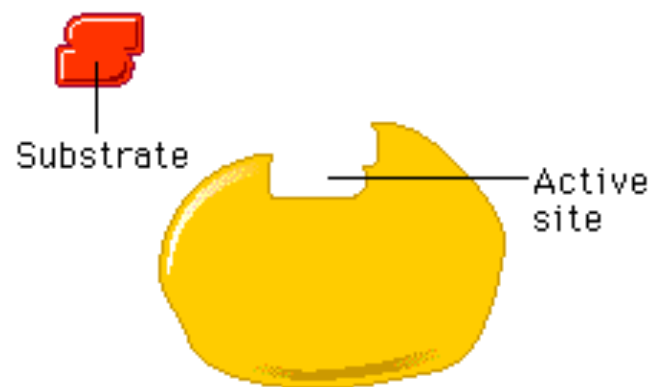
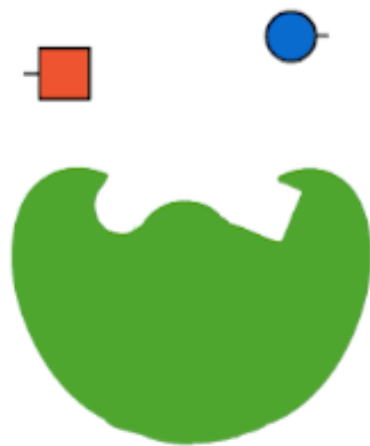
Figure 5: Limb Malformation in Infants Due to Thalidomide Consumption

50% of infants died
Rest were born with phocomelia

Applications of Stereochemistry : Drug Design

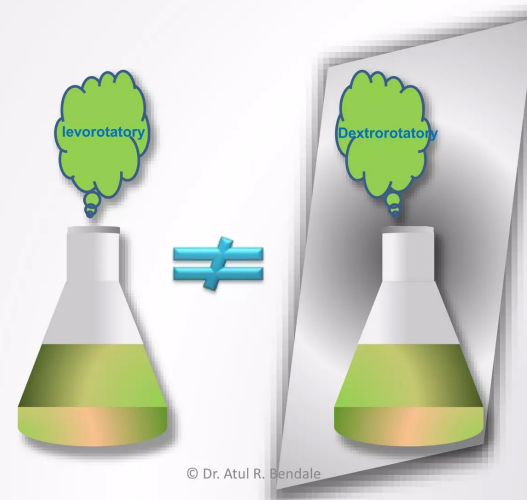
Enantiomers of a chiral drug have identical physical and chemical properties in an achiral environment. However, in a chiral environment, one enantiomer may display different chemical and pharmacologic behavior than the other enantiomer

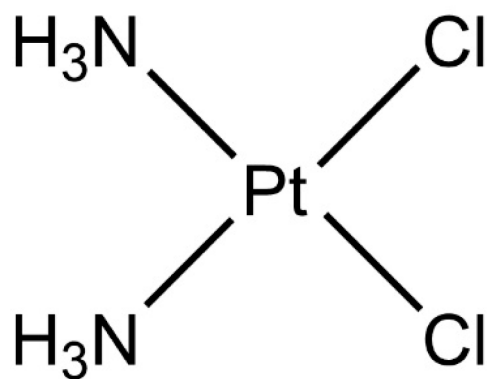
Living systems are chiral



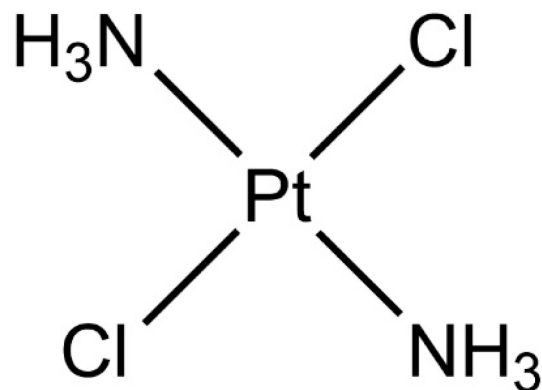
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- Enantiomers of a chiral drug may differ significantly in their bioavailability, rate of metabolism, metabolites, excretion, potency and selectivity for receptors, transporters and/or enzymes, and toxicity

Asymmetric Synthesis





(a) Cisplatin



(b) Transplatin

Cis isomer is clinically
used as an Anticancer
drug (chemotherapy)

Trans isomer is Inactive
Does not have any
anticancer effects